Skill – 1

M. LIKHITHA

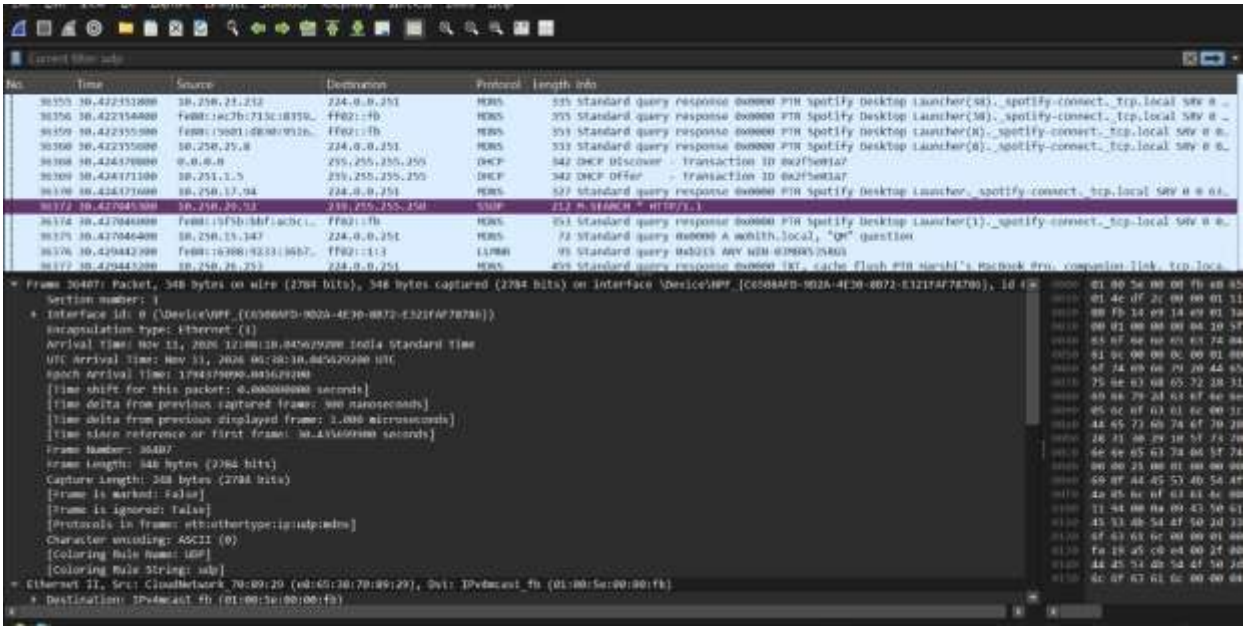
2520030575

STEP1: INSTALLATION OF WIRESHARK

STEP2: Select your network interface

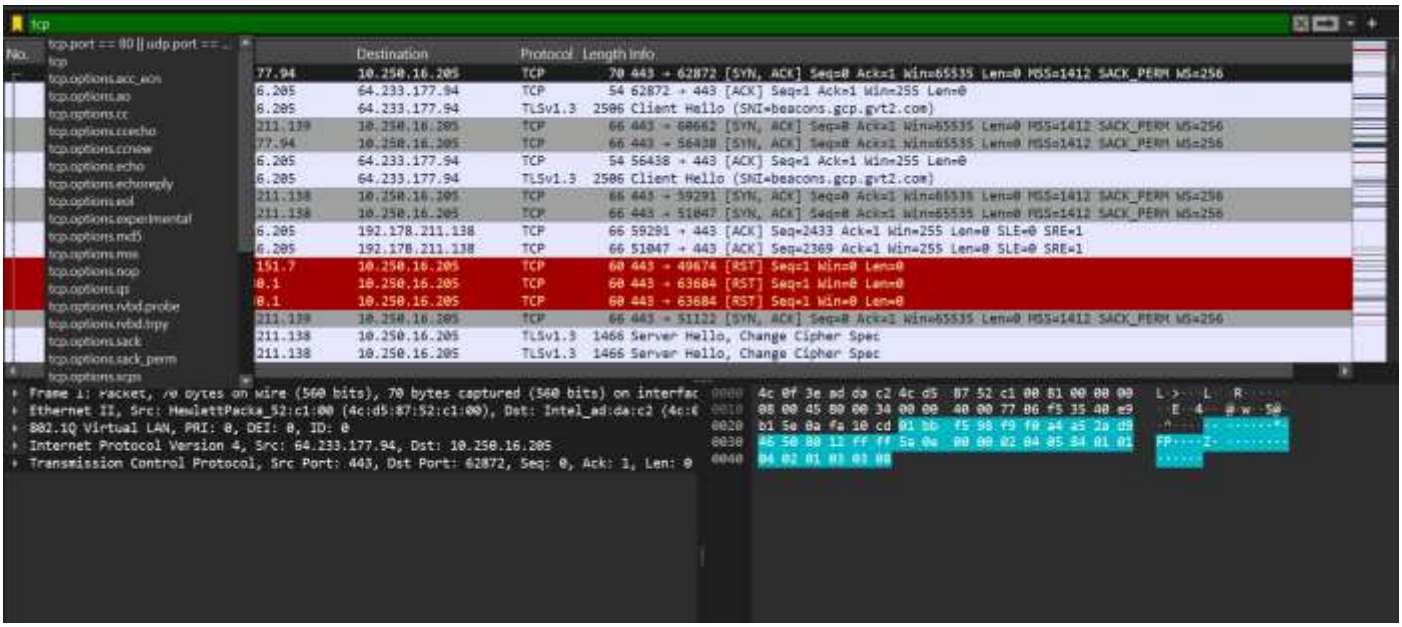
Connect through wifi then packet capture will start

STEP3: Generating network traffic and stoping it after browsing

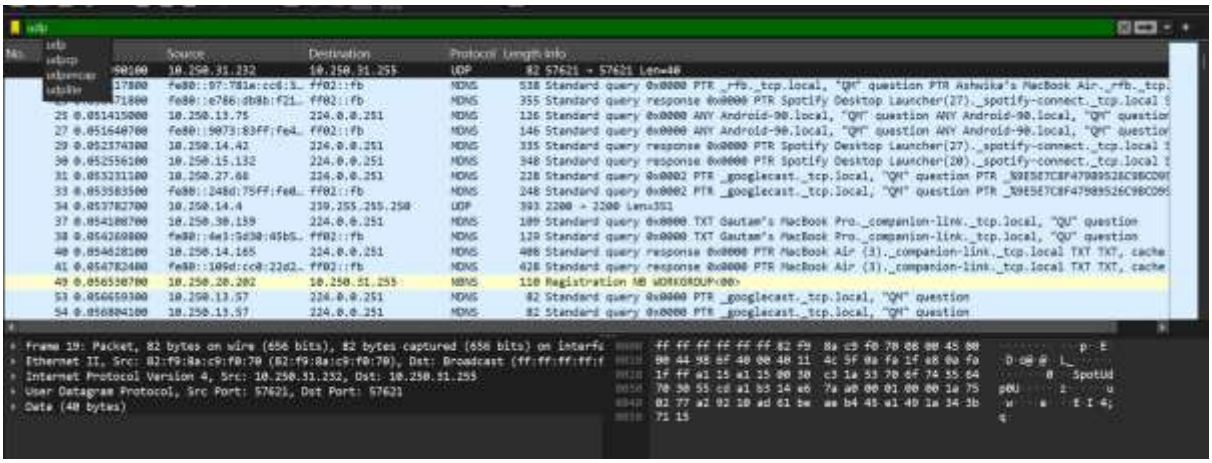


STEP4: Apply display filters In the Wireshark filter bar, enter each filter separately and press Enter

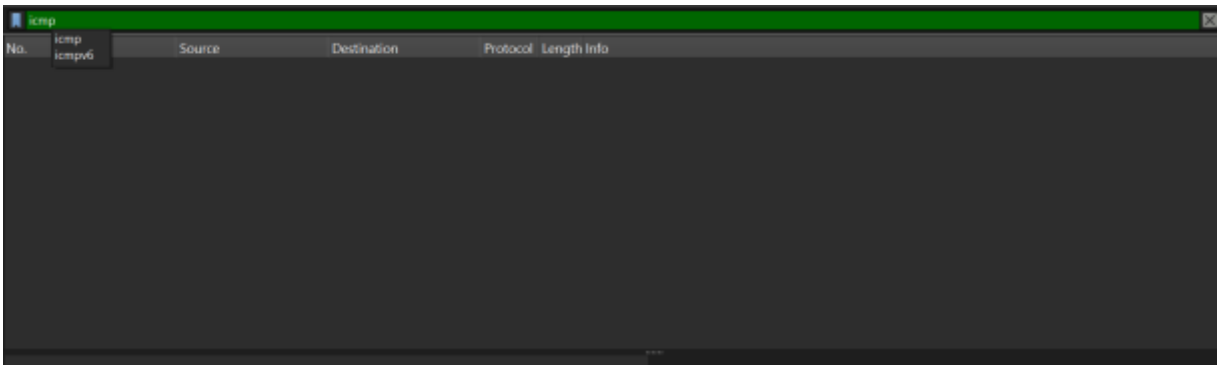
TCP



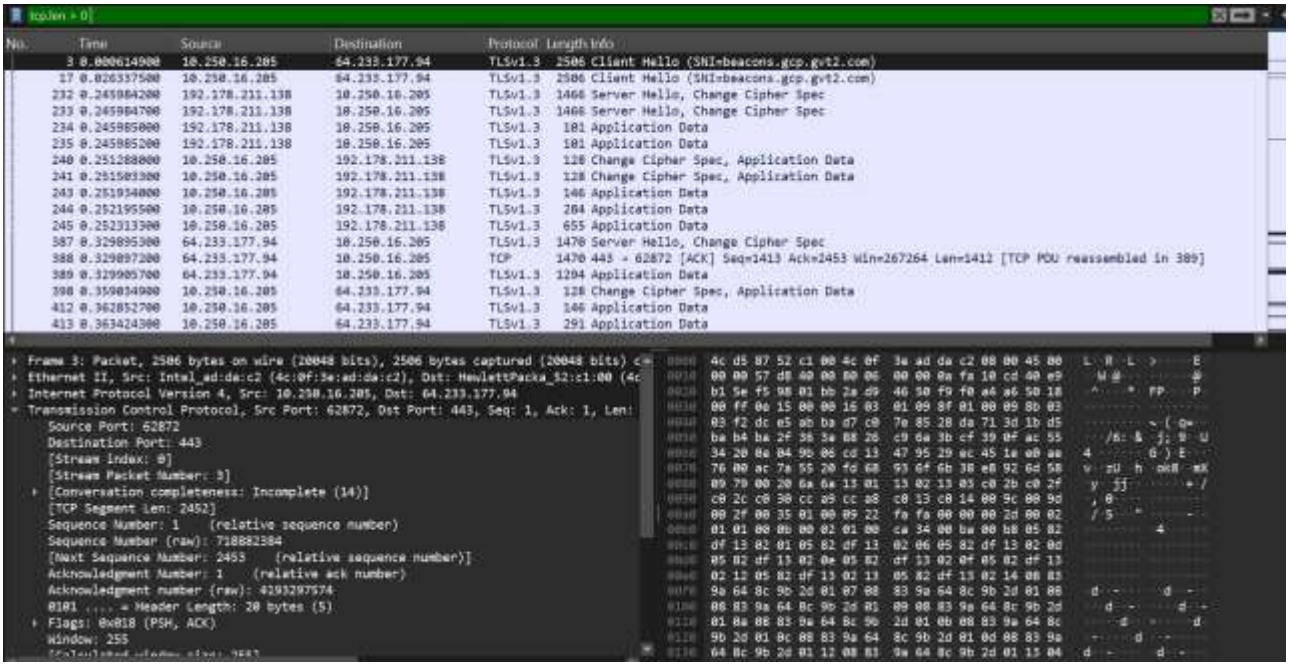
UDP



ICMP

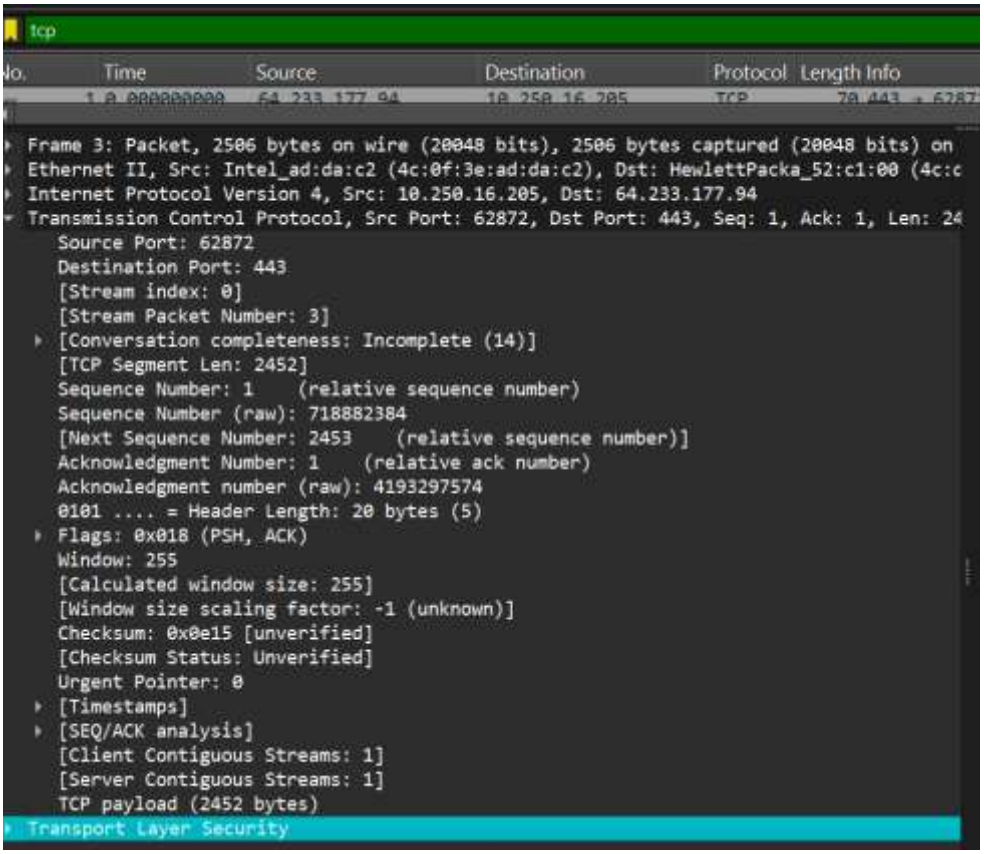


TCP.LEN>0

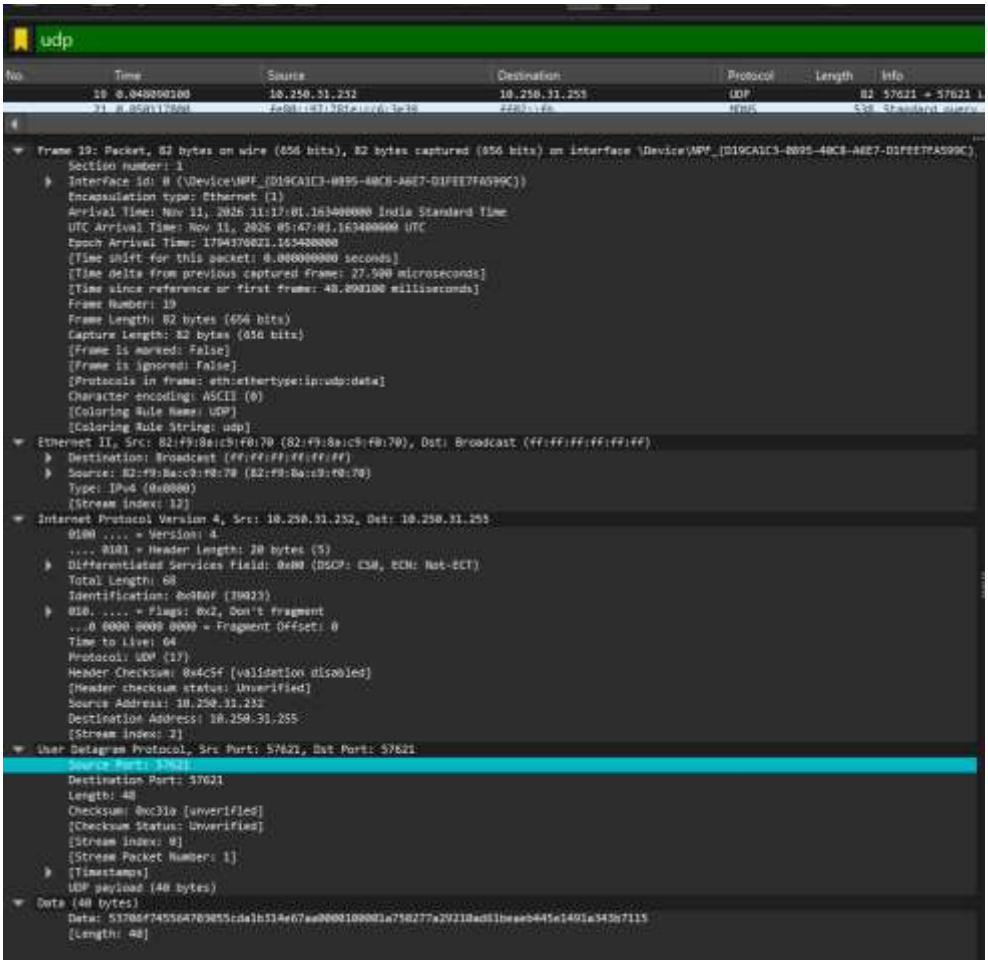


Step 5: Inspect packet headers

Select any TCP packet from the packet list



Select any UDP packet from the packet list



1. TCP packet calculation

Parameter	Observed value
Frame Length	2506 bytes
Ethernet Header	14 bytes
IPv4 Header	20 bytes
TCP Header	20 bytes
TCP Payload	2452 bytes
FCS	Not included in captured frame

Formula 1: Total packet size

$$\text{Total Size} = 14 + 20 + 20 + 2452$$

$$\boxed{2506 \text{ bytes}}$$

Formula 2: Total header overhead

$$\text{Overhead} = 14 + 20 + 20$$

$$\boxed{54 \text{ bytes}}$$

Payload efficiency

$$\text{Efficiency} = \frac{\text{Payload}}{\text{Total Frame Length}} \times 100$$

$$= \frac{2452}{2506} \times 100$$

$$\boxed{97.84\%}$$

UDP packet calculation

Parameter	Observed value
Frame Length	82 bytes
Ethernet Header	14 bytes
IPv4 Header	20 bytes
UDP Header	8 bytes

UDP Payload	40 bytes
FCS	Not included in captured frame

Formula 1: Total packet size

$$\text{Total Size} = 14 + 20 + 8 + 40$$

$$\boxed{82 \text{ bytes}}$$

Formula 2: Total header overhead

$$\text{Overhead} = 14 + 20 + 8$$

$$\boxed{42 \text{ bytes}}$$

Payload efficiency

$$\text{Efficiency} = \frac{\text{Payload}}{\text{Total Frame Length}} \times 100$$

$$= \frac{40}{82} \times 100$$

$$\boxed{48.78\%}$$

Comparison of TCP and UDP

Parameter	**TCP**	**UDP**
Frame Length	2506 bytes	82 bytes
Ethernet Header	14 bytes	14 bytes
IP Header	20 bytes	20 bytes
Transport Header	20 bytes	8 bytes
Payload	2452 bytes	40 bytes
Total Overhead	54 bytes	42 bytes
Payload Efficiency	97.84%	48.78%

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Application data = 1500 bytes

- FCS = 4 bytes

Total size:

$$14 + 20 + 20 + 1500 + 4 = 1558$$

Total overhead:

$$14 + 20 + 20 + 4 = 58$$

Payload efficiency:

$$\frac{1500}{1558} \times 100 = \boxed{96.28\%}$$